

KELOWNA, BC, Canada

WMO: 712030

Lat: 49.960N

Lon: 119.380W

Elev: 433

StdP: 96.23

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-16.9	-13.2	-20.8	0.6	-15.9	-17.2	0.9	-12.2	8.5	-1.6	7.3	-2.8	1.5	0	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	33.2	18.0	31.3	17.4	29.4	16.7	19.1	29.9	18.2	28.7	17.3	27.5	3.1	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	11.5	21.4	14.4	10.8	20.7	13.4	10.1	20.2	55.9	29.8	52.9	28.6	50.3	27.7	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	6.9	5.7	-20.8	36.3	4.8	1.5	-24.3	37.4	-27.1	38.3	-29.8	39.1	-33.3	40.2		
(5)			-21.2	21.3	4.8	1.3	-24.7	22.3	-27.5	23.0	-30.2	23.8	-33.7	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	-2.3	-1.2	3.8	8.1	12.9	16.4	20.2	19.3	14.1	7.2	1.8	-2.3	(6)
(7)		DBStd	8.83	5.00	4.61	3.85	3.06	3.39	3.11	2.92	3.02	3.25	3.32	4.43	4.73	(7)
(8)		HDD10.0	1704	380	314	193	73	11	1	0	0	4	98	247	382	(8)
(9)		HDD18.3	3843	638	547	450	308	171	73	15	26	131	346	497	641	(9)
(10)		CDD10.0	1053	0	0	1	15	101	194	316	288	128	11	0	0	(10)
(11)		CDD18.3	152	0	0	0	0	2	17	72	56	5	0	0	0	(11)
(12)		CDH23.3	2846	0	0	0	4	126	366	1228	968	155	1	0	0	(12)
(13)		CDH26.7	1122	0	0	0	0	28	124	542	391	37	0	0	0	(13)
(14)	Wind	WSAvg	1.9	1.7	1.7	2.0	2.2	2.2	2.2	1.9	1.7	1.6	1.6	1.6	(14)	
(15)	Precipitation	PrecAvg	466	47	31	30	29	46	57	31	28	31	42	45	50	(15)
(16)		PrecMax	583	74	56	46	58	97	106	112	69	75	73	93	81	(16)
(17)		PrecMin	334	27	18	15	11	16	27	0	2	4	15	18	23	(17)
(18)		PrecStd	65	12	9	7	12	19	22	24	18	19	15	18	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	9.8	17.1	23.4	29.8	33.4	36.1	34.8	30.4	20.2	13.0	8.4	(19)
(20)			MCWB	4.3	5.2	9.1	13.1	16.8	18.1	18.9	18.3	16.8	13.2	9.0	4.9	(20)
(21)		2%	DB	5.2	7.8	14.2	20.4	27.0	30.1	33.9	32.6	27.2	17.1	10.7	5.9	(21)
(22)			MCWB	3.1	4.1	7.5	11.2	15.2	16.7	18.1	17.9	16.0	11.3	7.4	3.3	(22)
(23)		5%	DB	4.0	6.0	12.3	17.8	24.1	27.4	32.0	30.9	24.9	15.2	9.1	4.2	(23)
(24)			MCWB	2.3	3.2	6.4	9.6	13.9	15.7	17.6	17.4	15.1	10.1	6.3	2.4	(24)
(25)		10%	DB	2.9	4.6	10.5	15.6	21.5	25.0	29.9	28.8	22.2	13.5	7.8	3.0	(25)
(26)			MCWB	1.7	2.2	5.5	8.4	12.7	14.8	17.0	16.6	13.8	9.2	5.6	1.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	6.2	9.7	13.7	17.5	19.2	20.6	20.3	18.4	14.1	10.0	5.6	(27)
(28)			MCD B	6.9	8.5	15.3	22.2	27.7	29.7	32.6	31.5	26.7	18.3	12.1	7.7	(28)
(29)		2%	WB	3.5	4.6	8.2	11.9	15.7	17.6	19.3	18.9	16.9	12.4	8.2	3.9	(29)
(30)			MCD B	4.9	6.8	13.1	18.8	24.5	28.1	30.6	29.5	25.2	15.6	9.8	5.3	(30)
(31)		5%	WB	2.5	3.5	7.1	10.3	14.6	16.5	18.3	18.0	15.7	10.9	6.9	2.6	(31)
(32)			MCD B	3.7	5.5	11.3	16.6	22.4	25.7	29.3	28.4	23.4	14.0	8.6	3.9	(32)
(33)		10%	WB	1.7	2.5	6.0	9.0	13.4	15.4	17.5	17.1	14.6	9.7	5.7	1.6	(33)
(34)			MCD B	2.7	4.3	9.8	14.5	20.1	23.3	28.0	27.2	21.2	12.9	7.3	2.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.6	8.3	10.8	13.5	14.4	14.3	16.7	17.0	14.8	11.2	7.4	5.9	(35)
(36)			MCD BR	6.5	8.9	13.5	18.0	19.9	19.8	21.4	21.0	19.6	13.7	8.9	6.7	(36)
(37)			MCWBR	4.8	6.1	8.1	10.2	9.8	8.6	8.4	8.7	9.6	8.3	6.1	5.0	(37)
(38)		5% WB	MCD BR	5.8	8.3	12.5	16.6	17.6	17.7	19.1	19.0	17.6	11.9	7.9	6.2	(38)
(39)			MCWBR	4.4	5.8	7.7	9.6	9.0	8.1	8.0	8.3	9.0	7.7	5.7	4.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.301	0.294	0.321	0.341	0.368	0.364	0.359	0.370	0.341	0.333	0.307	0.282	(40)	
(41)		taud	2.351	2.464	2.435	2.413	2.373	2.409	2.451	2.415	2.481	2.474	2.450	2.427	(41)	
(42)		Ebn at Noon	738	859	890	905	890	894	893	864	857	797	733	717	(42)	
(43)		Edh at Noon	66	76	93	106	115	112	106	105	88	75	60	53	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.93	1.90	3.07	4.51	5.39	5.68	6.49	5.54	3.87	2.18	0.99	0.71	(44)	
(45)		RadStd	0.10	0.19	0.31	0.35	0.43	0.53	0.52	0.38	0.34	0.28	0.12	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+120	+36
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page